Male sexuality and cardiovascular risk. A cohort study in patients with erectile dysfunction.

INTRODUCTION: Although penile blood flow (PBF) has been recommended as an additional diagnostic test in identifying erectile dysfunction (ED) patients at risk for latent cardiovascular disease, no study has ever assessed the possible association of PBF and the relational component of sexual function with incident major cardiovascular events (MACE). AIM: The aim of this study is to investigate whether severity of ED, PBF, and other factors related to a couple's relationship predict incident MACE. METHODS: A consecutive series of 1,687 patients was studied. Different clinical, biochemical, and instrumental (penile flow at color Doppler ultrasound) parameters were evaluated. MAIN OUTCOME MEASURES: Information on MACE was obtained through the City of Florence Registry Office. RESULTS: During a mean follow-up of 4.3 +/- 2.6 years, 139 MACE, 15 of which were fatal, were observed. Cox regression analysis, after adjustment for age and Chronic Disease Score, showed that severe ED predicted MACE (hazard ratio [HR] 1.75; 95% confidence interval 1.10-2.78; $P < 0.05$). In addition, lower PBF, evaluated both in flaccid (before) and dynamic (after prostaglandin-E1 stimulation) conditions, was associated with an increased risk of MACE (HR = 2.67 [1.42-5.04] and 1.57 [1.01-2.47], respectively, for flaccid [<13 cm/second] and dynamic [<25 cm/second] peak systolic velocity; both $P < 0.05$). Reported high sexual interest in the partner and low sexual interest in the patient proved to have a protective effect against MACE. CONCLUSIONS: The investigation of male sexuality, and in particular PBF, and sexual desire, could provide insights not only into present cardiovascular status but also into prospective risk.